

Name: \_\_\_\_\_

Date: \_\_\_\_\_



# EARLY ALGEBRA



## LEVEL 3: SYMBOLS

### 1 Find the value of each symbol.

Use the equations to find the value of each shape.

$$\square + \square = 10 \quad \square = \square$$

$$\bigcirc + \bigcirc + \bigcirc = 15 \quad \bigcirc = \square$$

$$\triangle + \bigcirc = 9 \quad \triangle = \square$$

$$\star + \triangle = 11 \quad \star = \square$$

### 2 Fill in the missing numbers.

Replace each symbol with the correct number to make the equation true.

$$\heartsuit + 4 = 10 \quad \heartsuit = \square$$

$$\diamondsuit - 3 = 5 \quad \diamondsuit = \square$$

$$\bullet + 7 = 13 \quad \bullet = \square$$

$$\blacktriangle + 6 = 12 \quad \blacktriangle = \square$$

### 3 Picture equations.

Write a number sentence for each picture using a symbol, then solve it.

$$\text{apple} + \text{apple} + \text{apple} = 12$$

Let  $\text{apple} = \square$        $\square + \square = \square$

$$\star + \star + \star + \star = 16$$

Let  $\star = \square$        $\square + \square + \square = \square$

$$\text{butterfly} + \text{butterfly} + \text{butterfly} - \text{butterfly} = 8$$

Let  $\text{butterfly} = \square$        $\square - \square = \square$

$$\text{flower} + \text{flower} + \text{flower} = 20$$

Let  $\text{flower} = \square$        $\square + \square = \square$

### 4 Match the expressions.

Draw a line to match each expression on the left with an equivalent expression on the right.

A.  $\diamondsuit + 5$

1.  $6 + \blacktriangle$

B.  $7 + \bullet$

2.  $3 + \star$

C.  $\blacktriangle + 6$

3.  $\bullet + 7$

D.  $\star + 3$

4.  $5 + \diamondsuit$

### ★ CHALLENGE! ★

If  $\star = 2$ ,  $\bullet = 5$  and  $\blacktriangle = 8$ ,  
what is the value of  $\star + \bullet + \blacktriangle$ ?

